Undergraduate Curriculum

The new Department of Physics Undergraduate Curriculum aims for students to determine their own choice of junior and senior tracks after getting fundamental freshmen and sophomore courses. Selective courses can be chosen either from preparatory courses for physics graduate programs or from other disciplines for interdisciplinary graduate programs.

One of the main purposes of this new curriculum is to ready the students for research by project courses; also, through summer internship, students can gain experience at research companies and institutions as well as research groups.

First Year

Fall Semester			Spring Semester		
Code	Title	Credit	Code	Title	Credit
PHYS 101	General Physics I	4	**PHYS 102**	General Physics II	4
MATH 101	Calculus I	4	**MATH 102**	Calculus II	4
CHEM 101	Principles of Chemistry I	4	**CS 113**	Introduction to Computing in Engineering for Engineers	4
ENG 101	English and Composition I	3	**ENG 102**	English and Composition II	3

PHYS 120	Orientation for Physics Majors I	1	**PHYS 124**	Freshman Project	2
TURK 101	Turkish I	2	**TURK 102**	Turkish II	2
GE 100	Orientation	1			

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Second Year

Fall Semester

Code	Title	Credit
MBG 105	Principles of Biology	3
PHYS 211	Waves, Fluids and Thermodynamics	4

Spring Semester

Code	Title	Credit
PHYS 242	Advanced Calculus for Applications in Physics	3
PHYS 212	Optics and Modern Physics	4

MATH 241	Engineering Mathematics I	4	**MATH 242**	Engineering Mathematics II	3
CHEM 201	Materials Science and Technology	3	**PHYS 218**	Analytical Mechanics	3
HIST 200	History of Turkey	4		***Elective***	3
GE 250	Collegiate Activities Program I	–	**GE 251**	Collegiate Activities Program II	1

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PHYS 291 Summer Practice (non-credit)

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Third Year

Fall Semester			Spring Semester		
Code	Title	Credit	Code	Title	Credit

PHYS 315	Electromagnetic Theory I	3
PHYS 325	Quantum Mechanics I	3
PHYS 334	Statistical Physics	3
	Technical Elective	3
HUM 111	Cultures, Civilizations and Ideas I	3
PHYS 291	Summer Practise	–

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Fourth Year

Fall Semester

PHYS 371	Numerical Methods in Physics	3
PHYS 374	Experimental Methods of Physics	4
	Physics Elective	3
	Elective	3
HUM 112	Cultures, Civilizations and Ideas II	3

Spring Semester

Code	Title	Credit	Code	Title	Credit
PHYS 491	Senior Project	3	**PHYS 492**	Senior Project II	3
	Physics Elective	3		***Technical Elective***	3
	Technical Elective	3		***Technical Elective***	3
	Elective	3		***Elective***	3
	Non-Technical Elective	3		***Non-Technical Elective***	3

Total # Credits: 135

Physics Elective

Any course offered by the Physics department with a 3XX or 4XX code (third year or fourth year course.)

Sample Tracks

Below are several example tracks that can be formed by students by ***their*** own choice. These are mere samples, different tracks can be formed.

Theoretical Physics

Third Year

Fall Semester

Code	Title	Credit
PHYS 315	Electromagnetic Theory I	3
PHYS 325	Quantum Mechanics I	3
PHYS 334	Statistical Physics	3

Spring Semester

Code	Title	Credit
PHYS 371	Numerical Methods in Physics	3
PHYS 374	Experimental Methods of Physics	3
PHYS 316	Electromagnetic Theory II	3

	Technical Elective	3	**PHYS 326**	Quantum Mechanics II	3
HUM 111	Cultures, Civilizations and Ideas I	3	**HUM 112**	Cultures, Civilizations and Ideas II	3

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Fourth Year

Fall Semester

Code	Title	Credit
PHYS 491	Senior Project	3
PHYS 445	Condensed Matter Physics I	3
PHYS 449	Group Theory	3

Spring Semester

Code	Title	Credit
PHYS 492	Senior Project II	3
PHYS 446	Condensed Matter Physics II	3
PHYS 438	Atomic, Molecular and Optical Physics	3

PHYS 478	Theory of Relativity	3	**PHYS 453**	Nuclear and Particle Physics	3
	Non-Technical Elective	3		***Non-Technical Elective***	3

Condensed Matter Physics

Third Year

Fall Semester

Code	Title	Credit
PHYS 315	Electromagnetic Theory I	3
PHYS 325	Quantum Mechanics I	3
PHYS 334	Statistical Physics	3

Spring Semester

Code	Title	Credit
PHYS 371	Numerical Methods in Physics	3
PHYS 374	Experimental Methods of Physics	3
PHYS 316	Electromagnetic Theory II	3

	Technical Elective	3	**PHYS 326**	Quantum Mechanics II	3
HUM 111	Cultures, Civilizations and Ideas I	3	**HUM 112**	Cultures, Civilizations and Ideas II	3

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Fourth Year

Fall Semester

Code	Title	Credit
PHYS 491	Senior Project	3
PHYS 445	Condensed Matter Physics I	3
PHYS 449	Group Theory	3

Spring Semester

Code	Title	Credit
PHYS 492	Senior Project II	3
PHYS 446	Condensed Matter Physics II	3
PHYS 438	Atomic, Molecular and Optical Physics	3

PHYS 420	Nanoscience and Nanotechnology I	3	**PHYS 475**	Semiconductor Device Physics	3
	Non-Technical Elective	3		***Non-Technical Elective***	3

Photonics

Third Year

Fall Semester

Code	Title	Credit
PHYS 315	Electromagnetic Theory I	3
PHYS 325	Quantum Mechanics I	3
PHYS 334	Statistical Physics	3

Spring Semester

Code	Title	Credit
PHYS 371	Numerical Methods in Physics	3
PHYS 374	Experimental Methods of Physics	3
PHYS 316	Electromagnetic Theory II	3

	Technical Elective	3	**PHYS 326**	Quantum Mechanics II	3
HUM 111	Cultures, Civilizations and Ideas I	3	**HUM 112**	Cultures, Civilizations and Ideas II	3

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Fourth Year

Fall Semester

Code	Title	Credit
PHYS 491	Senior Project	3
PHYS 445	Condensed Matter Physics I	3
PHYS 415	Photonics	3

Spring Semester

Code	Title	Credit
PHYS 492	Senior Project II	3
PHYS 477	Lasers	3
PHYS 438	Atomic, Molecular and Optical Physics	3

EEE 211	Analog Electronics	3	**PHYS 475**	Semiconductor Device Physics	3
	Non-Technical Elective	3		***Non-Technical Elective***	3

Nanoscience and Nanotechnology

Third Year

Fall Semester

Code	Title	Credit
PHYS 315	Electromagnetic Theory I	3
PHYS 325	Quantum Mechanics I	3
PHYS 334	Statistical Physics	3

Spring Semester

Code	Title	Credit
PHYS 371	Numerical Methods in Physics	3
PHYS 374	Experimental Methods of Physics	3
PHYS 316	Electromagnetic Theory II	3

	Technical Elective	3	**PHYS 326**	Quantum Mechanics II	3
HUM 111	Cultures, Civilizations and Ideas I	3	**HUM 112**	Cultures, Civilizations and Ideas II	3

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Fourth Year

Fall Semester

Code	Title	Credit
PHYS 491	Senior Project	3
PHYS 445	Condensed Matter Physics I	3
PHYS 420	Nanoscience and Nanotechnology I	3

Spring Semester

Code	Title	Credit
PHYS 492	Senior Project II	3
PHYS 438	Atomic, Molecular and Optical Physics	3
PHYS 477	Lasers	3

EEE 211	Analog Electronics	3	**PHYS 475**	Semiconductor Device Physics	3
	Non-Technical Elective	3		***Non-Technical Elective***	3

Biophysics

Third Year

Fall Semester

Code	Title	Credit
PHYS 315	Electromagnetic Theory I	3
PHYS 325	Quantum Mechanics I	3
PHYS 334	Statistical Physics	3

Spring Semester

Code	Title	Credit
PHYS 371	Numerical Methods in Physics	3
PHYS 374	Experimental Methods of Physics	3
MBG 338	Microbiology (or) Principles of Organic Chemistry	3

			CHEM 233		
MBG 311	Biochemistry I	3	**MBG 312**	Biochemistry II	3
HUM 111	Cultures, Civilizations and Ideas I	3	**HUM 112**	Cultures, Civilizations and Ideas II	3

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Fourth Year

Fall Semester			Spring Semester		
Code	Title	Credit	Code	Title	Credit
PHYS 491	Senior Project	3	**PHYS 492**	Senior Project II	3
PHYS 445	Condensed Matter Physics I	3	**MBG 316**	Physiology	3

PHYS 415	Photonics	3	**MBG 473**	Biomolecules, Biomaterials and Bioprocesses	3
MBG 326	Introduction to Bioinformatics	3	**MBG 474**	Introduction to Computational Biology	3
	Non-Technical Elective	3		***Non-Technical Elective***	3

Biomedical

Third Year

Fall Semester

Code	Title	Credit
PHYS 315	Electromagnetic Theory I	3

Spring Semester

Code	Title	Credit
PHYS 371	Numerical Methods in Physics	3

PHYS 325	Quantum Mechanics I	3	**PHYS 374**	Experimental Methods of Physics	3
PHYS 334	Statistical Physics	3	**MBG 338**CHEM 233	Microbiology (or)Principles of Organic Chemistry	3
MBG 311	Biochemistry I	3	**MBG 312**	Biochemistry II	3
HUM 111	Cultures, Civilizations and Ideas I	3	**HUM 112**	Cultures, Civilizations and Ideas II	3

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Fourth Year

Fall Semester

Spring Semester

Code	Title	Credit	Code	Title	Credit

PHYS 491	Senior Project	3	**PHYS 492**	Senior Project II	3
PHYS 415	Photonics	3	**MBG 316**	Physiology	3
MBG 222	Fundamentals of Molecular Genetics	3	**PSYC 405**	Introduction to FMRI	3
MBG 326	Introduction to Bioinformatics	3	**PSYC 432**	Cognitive Neuroscience	3
	Non-Technical Elective	3		***Non-Technical Elective***	3

Chemical Physics

Second Year

Fall Semester

Code	Title	Credit

Spring Semester

Code	Title	Credit

MBG 105	Principles of Biology	3	**PHYS 242**	Advanced Calculus for Applications in Physics	4
PHYS 211	Waves, Fluids and Thermodynamics	4	**PHYS 212**	Optics and Modern Physics	4
MATH 241	Engineering Mathematics I	4	**MATH 242**	Engineering Mathematics II	4
CHEM 201	Materials Science and Technology	3	**PHYS 218**	Analytical Mechanics	3
HIST 200	History of Turkey	4	**CHEM 102**	Principles of Chemistry II (+L)	3
GE 250	Collegiate Activities Program I	–	**GE 251**	Collegiate Activities Program II	1
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PHYS 291 Summer Practice (non-credit)

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Third Year

Fall Semester

Code	Title	Credit
PHYS 315	Electromagnetic Theory I	3
PHYS 325	Quantum Mechanics I	3
PHYS 334	Statistical Physics	3
CHEM 211	Analytical Chemistry I	3
HUM 111	Cultures, Civilizations and Ideas I	3

Spring Semester

Code	Title	Credit
PHYS 371	Numerical Methods in Physics	3
PHYS 374	Experimental Methods of Physics	3
PHYS 326	Quantum Mechanics II	3
CHEM 212	Analytical Chemistry II	3
HUM 112	Cultures, Civilizations and Ideas II	3

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Fourth Year

Fall Semester			Spring Semester		
Code	Title	Credit	Code	Title	Credit
PHYS 491	Senior Project	3	**PHYS 492**	Senior Project II	3
PHYS 420	Nanoscience and Nanotechnology I	3	**PHYS 438**	Atomic, Molecular and Optical Physics	3
CHEM 323	Physical Chemistry I	3	**CHEM 323**	Physical Chemistry I	3
CHEM 341	Inorganic Chemistry I	3	**CHEM 342**	Inorganic Chemistry II	3
	Non-Technical Elective	3		***Non-Technical Elective***	3

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